

How the out-of-hours review is calculated

Every formula in the engine, worked through with the numbers from a real building, so any figure in a client report can be reproduced by hand and defended in a conversation.

Worked example throughout: **Hog office building**, 18,213 m², 8,760 hourly readings for calendar year 2017, from the Building Data Genome Project 2 open dataset. Tariff 0.110 per kWh. Stated hours Mon–Fri 08:00–18:00, plus 9 closed dates.

STEP 1

Reading the file

Interval length

The gap between consecutive timestamps is taken, and the *modal* gap — the one that occurs most often — becomes the interval. The mode is used rather than the mean so that a few missing rows cannot stretch the estimate.

$$\Delta t = \text{mode}(t_i - t_{i-1}) \quad h = \Delta t / 60 \quad [\text{hours per reading}]$$

Hog: $\Delta t = 60 \text{ min} \rightarrow h = 1.0 \text{ h}$

A quarter-hourly German Lastgang would give $\Delta t = 15 \text{ min} \rightarrow h = 0.25 \text{ h}$.

Energy and power

A meter export gives either energy per interval (kWh) or average power over the interval (kW). They are not the same number and confusing them scales every result by 4 on quarter-hourly data. The unit is declared per client, and the conversion is one line:

$$\begin{array}{ll} \text{if the file is kWh:} & E_i = \text{value}_i \quad P_i = E_i / h \\ \text{if the file is kW:} & P_i = \text{value}_i \quad E_i = P_i \times h \end{array}$$

Hog, first reading: $E = 66.05 \text{ kWh}$ over $1.0 \text{ h} \rightarrow P = 66.05 \text{ kW}$. With $h = 1$ the two are numerically equal, which is exactly why an hourly file hides the mistake and a quarter-hourly one does not.

Gaps are never filled

A complete calendar of the period at the interval length is built, and the readings present are subtracted from it. Missing intervals stay missing. Filling them — with a mean, a previous value, anything — would change the client's total and make the report impossible to reconcile against their own file.

$$\text{coverage} = n_{\text{present}} / n_{\text{expected}}$$

Hog: 8,760 present, 8,760 expected → coverage **100.00%**, 0 gaps. The synthetic test fixture has 34,905 of 35,040 → 99.61%, 3 gaps.

STEP 2

Splitting closed from occupied

This is the only place the client's own information enters the arithmetic, and everything downstream depends on it. A reading is **occupied** when all four conditions hold; otherwise it is **closed**.

$$\text{occupied}_i = \text{weekday}(t_i) \in W \wedge h_{\text{open}} \leq \text{hour}(t_i) < h_{\text{close}} \wedge \text{date}(t_i) \notin \text{Holidays} \\ \wedge \text{date}(t_i) \notin \text{Shutdowns}$$

$$\text{closed}_i = \neg \text{occupied}_i$$

Hog: $W = \{\text{Mon...Fri}\}$, $h_{\text{open}} = 8$, $h_{\text{close}} = 18$, 9 holiday dates.

$n_{\text{closed}} = \mathbf{6,250}$ readings $n_{\text{occupied}} = \mathbf{2,510}$ ($6,250 + 2,510 = 8,760$ ✓)

Closed hours = $6,250 \times 1.0 = \mathbf{6,250}$ h of the year's 8,760.

The partition is exhaustive and exclusive by construction — no reading can be counted twice or dropped. Check 3 in the report verifies this holds in the delivered numbers, not just in theory.

Energy either side of the split

$$E_{\text{closed}} = \sum E_i \text{ where } \text{closed}_i \quad E_{\text{total}} = \sum E_i$$

$$E_{\text{total}} = \mathbf{1,219,928.85 \text{ kWh}}$$

$$E_{\text{closed}} = \mathbf{649,634.76 \text{ kWh}}$$

$$E_{\text{open}} = 570,294.09 \text{ kWh} \quad (\text{sums back exactly } \checkmark)$$

$$\text{closed share} = 649,634.76 / 1,219,928.85 = \mathbf{53.25\%}$$

$$\text{cost of closed energy} = 649,634.76 \times 0.110 = \mathbf{71,459.82}$$

MEASURED These are sums of the client's own readings. Nothing is modelled.

STEP 3

The load while closed

The *median* of the closed-period power readings, not the mean. A median is unmoved by a handful of extreme intervals — a one-off test run, a meter spike, a Saturday event — which a mean would drag upwards and overstate the opportunity.

$$\tilde{P}_{\text{closed}} = \text{median}\{ P_i : \text{closed}_i \} \quad \bar{P}_{\text{open}} = \text{mean}\{ P_i : \text{occupied}_i \}$$

Hog: $\tilde{P}_{\text{closed}} = 71.10 \text{ kW}$ $\bar{P}_{\text{open}} = 227.21 \text{ kW}$
ratio = $71.10 / 227.21 = 31\%$ — roughly a third of the working-day load is still running with nobody in the building.

This ratio is also the qualification test. Above 55% the building either runs 24/7 or the stated hours are wrong, and the meter alone cannot tell the two apart. The engine refuses to issue a report in that case rather than publish a savings figure that is really a guess. The second real building tested (Eagle office) came out at 86% and was correctly rejected.

STEP 4

The level the building has already reached

The report needs a reference point for "how low could this go" that is not an industry benchmark — because a benchmark invites the answer "*our building is different*", and it would be an assumption. Instead the reference comes from the building's own record.

For each calendar day, the minimum power reached during that day's closed periods is taken. That gives one number per day — the quietest moment of each night. The 10th percentile across those daily minima is the reference level.

$$m_d = \min\{ P_i : \text{closed}_i \wedge \text{date}(t_i) = d \}$$
$$P_{\text{floor}} = \text{percentile}_{10}(\{ m_d \})$$

Hog: 365 daily minima $\rightarrow P_{\text{floor}} = 47.72 \text{ kW}$
Nights at or within 5% of that level: **63** — so the site reached it on about one night in six. It is demonstrably achievable with the equipment already installed.

The 10th percentile rather than the outright minimum: a single lowest reading could be a meter dropout or a one-off shutdown, and one freak value should not become the target.

What holding that level all year would have been worth

$$E_{\text{avoidable}} = \sum \max(P_i - P_{\text{floor}}, 0) \times h \quad \text{over closed intervals}$$

Hog: **351,854.81 kWh** = **38,704.03** at 0.110/kWh (28.8% of the year's consumption)

Known limitation, stated in the report. This method under-reports when waste is *constant*. A building that is badly run every single night has no good night to compare against, so its own floor is also bad. It is therefore a lower bound on the opportunity, never an upper one — which is why the report leads with the measured closed-period share instead, and presents this figure as supporting evidence.

STEP 5

What one kilowatt is worth

The headline commercial number, and deliberately the simplest thing in the document. No target is assumed and no saving is promised — it is a conversion rate the client applies to whatever reduction they decide to go after.

$$\text{value of 1 kW removed} = \text{closed hours} \times \text{tariff}$$

Hog: $6,250 \text{ h} \times 0.110 \text{ /kWh} = \mathbf{687.50 \text{ per kW per year}}$

Which drives the scenario table directly:

10% of 71.10 kW = 7.11 kW → 44,437 kWh → **4,888/yr**

25% of 71.10 kW = 17.77 kW → 111,091 kWh → **12,220/yr**

50% of 71.10 kW = 35.55 kW → 222,183 kWh → **24,440/yr**

DERIVED Arithmetic on the client's own readings and their own tariff. A facilities manager can check it on a phone.

STEP 6

When the building actually switches on and off

The median power at each time of day is taken across working days only, giving a single representative daily profile. "Something switched on" is defined as that profile lifting 10% of its daily range above its overnight floor.

$$\text{profile}(\tau) = \text{median}\{ P_i : \text{time}(t_i) = \tau, \text{ weekday} \in W \}$$

$$\text{threshold} = \min(\text{profile}) + 0.10 \times (\max(\text{profile}) - \min(\text{profile}))$$

$$\tau_{\text{on}} = \min\{ \tau : \text{profile}(\tau) \geq \text{threshold} \} \quad \tau_{\text{off}} = \max\{ \tau : \text{profile}(\tau) \geq \text{threshold} \}$$

Hog: min 62.99 kW, max 243.34 kW

threshold = $62.99 + 0.10 \times (243.34 - 62.99) = 81.03 \text{ kW}$

$\tau_{\text{on}} = 08:00$ (no early start here) $\tau_{\text{off}} = 22:00 \rightarrow 4.0 \text{ hours}$ of running past the stated 18:00 close.

10% and not 25%: a higher threshold reports the middle of the morning ramp rather than its beginning, which is not the question the client is asking. This was found by testing against the synthetic fixture, where the true start was known to be 05:30 and a 25% threshold reported 06:15.

STEP 7

The remaining measures

FIGURE	FORMULA	HOG
Peak demand	$\max(P_i)$	314.39 kW 24 Oct, 14:00
Weekend load	$\text{mean}\{P_i : \text{Sat v Sun}\}$	94.32 kW
Energy on stated-empty days	ΣE_i over holiday and shutdown dates	24,852.62 kWh 9 days · 2,734
Energy use intensity	$E_{\text{total}} / \text{floor area}$	66.98 kWh/m ² ·yr

STEP 8

The nine checks, and what each one catches

Every check runs on every job. **If any fails, no report is written.** This is a hard gate in the code, not a warning.

#	CHECK	WHAT IT CATCHES
1	Report total = sum of rows used	any arithmetic that lost or duplicated energy
2	Rows used reconcile to the raw file	rows silently dropped during parsing
3	Closed + occupied = total	an interval counted twice or missed by the split
4	No negative readings	export or sign errors; an on-site generator
5	Coverage $\geq 95\%$	too much missing data to make claims about a year
6	Floor < typical closed load	a degenerate percentile on odd data
7	Avoidable $\subseteq$ closed-period energy	the saving exceeding what was ever used
8	Stated hours fit the pattern (<55%)	a 24/7 site, or wrong hours in the job file
9	Avoidable < 40% of annual use	an implausible headline slipping through

Separately, a companion Excel workbook re-derives eleven of the report's figures using ordinary spreadsheet formulas — SUMIF, COUNTIF, AVERAGEIF, MEDIAN — reading only the raw timestamp and kWh columns. The closed/occupied flag is recomputed there too, with WEEKDAY and HOUR, so not even the split is taken on trust. On the worked example every difference is exactly zero.

STEP 9

What this method deliberately does not do

Saying so plainly is what keeps the rest defensible.

- **It does not identify equipment.** The meter says how much and when, never what. Naming the circuit needs a clamp meter or sub-metering.
- **It does not weather-correct.** A cold winter raises consumption and this method does not adjust for it. That is acceptable here because the closed/occupied comparison is within the same year and the same weather, but it means the annual total is not comparable across years without correction.
- **It does not promise savings.** It converts a reduction the client chooses into money. What is achievable depends on plant that has not been inspected.
- **It does not model, forecast or optimise.** Every figure is a sum, a median, a percentile or a product — nothing is fitted or predicted.
- **It carries no compliance status.** It is not an energy audit under EN 16247 or DIN EN ISO 50001, and does not certify anything.

Method document for internal use and for answering client questions. Worked figures are from an open research dataset, not from a client. Building Data Genome Project 2, building Hog_office_Shawna, 2017.